Lab 1: Data Loading and Environment Setup

Learning Objectives

By completing this lab, you will:

- Upload data files to Splunk Enterprise
- Configure source types for different data formats
- Set host field values for data identification
- Understand the data ingestion workflow
- Verify successful data ingestion

Prerequisites

- A running Splunk instance — see **Lab Environment Setup** below if you do not have one
- Administrative access to Splunk Web interface
- Lab data files (already provided in data folder)

> **Data window**: The course dataset covers roughly **26 days** ending at the moment `labs/data/generate_course_data.py` was last run — deliberately sized to sit inside Splunk's **Last 30 days** preset. Set the time picker to **Last 30 days** for every exercise unless a step says otherwise.
>
> Regenerate the data within **4 days** of delivering the class; past that the preset starts clipping the oldest events and the answer-key counts drift.

Lab Environment

- **Splunk Version:** Enterprise (latest stable)
- **User Account:** admin
- **Password:** password
- **Default Index:** main

Lab Environment Setup

If your instructor has already provided a running Splunk instance, confirm you can reach it with **Step 3** below and then continue to Task 1. Otherwise, set one up using **Option A** (works with no network) or **Option B** (needs internet).

Step 1: Start Splunk

Option A: Splunk in Docker (recommended; works offline)

1. Confirm Docker is running:

   ```
   docker info
   ```

2. Confirm the Splunk image is already on this machine:

   ```
   docker images splunk/splunk
   ```

   If that prints nothing, pull it **while you still have internet** (~6.5 GB, so do not leave this until class time):

   ```
   docker pull --platform linux/amd64 splunk/splunk:latest
   ```

3. Start the container:

   ```
   docker run -d -p 8000:8000 -e SPLUNK_GENERAL_TERMS=--accept-sgt-current-at-splunk-com -e
           SPLUNK_START_ARGS=--accept-license --platform linux/amd64 -e
           "SPLUNK_PASSWORD=password" --name splunk splunk/splunk:latest
   ```

   If a `splunk` container already exists from a previous session, start that one instead of creating a second:

   ```
   docker start splunk
   ```

4. Splunk takes a few minutes to come up on first start. Wait until the status column reads `healthy`:

   ```
   docker ps --filter name=splunk
   ```

Option B: Splunk Cloud trial (requires internet)

Request a free trial instance at https://www.splunk.com/en_us/download/splunk-cloud.html. You will choose your own password during signup; use that instead of the one listed above.

> **Not suitable for an offline classroom.** Splunk Cloud is hosted, so every search needs a live connection. Use Option A if the room may be offline.

Step 2: Confirm the data files are present

The three log files live in the course materials under `labs/data/`. Task 1 walks through verifying them in detail.

Step 3: Verify access

1. Open `http://localhost:8000` (Docker) or your Splunk Cloud URL
2. Log in — for Docker, username `admin` and password `password`
3. Confirm **Settings** appears in the top menu, which indicates admin rights

Checkpoint: Splunk Web loads and you are logged in as an administrator.

Scenario

You have recently joined the team at Buttercup Games as a Splunk Administrator. Your first assignment is to ingest three critical data sources into your Splunk Enterprise instance for security monitoring and analysis:

1. Web application access logs
2. Linux secure logs from web servers
3. Database audit logs

These datasets cover roughly 26 days of historical data that will be used for training and analysis purposes.

Task 1: Verify Lab Data Files

Overview

The lab data files are already provided in your course materials folder for this exercise.

Steps:

1. **Locate the lab data directory:** Navigate to the `data` folder in your course materials directory
2. **Verify the following files are present:**
   - `access_30DAY.log` (Web application logs)
   - `linux_s_30DAY.log` (Linux secure logs)
   - `db_audit_30DAY.csv` (Database audit logs)
   - `products.csv` (Product lookup data for later labs)
3. **Note about the data:**
   - These files contain roughly 26 days of historical data
   - The data is static (not real-time) for training purposes
   - You will use these same files throughout the course

Checkpoint ✓

- [ ] Located the course data directory
- [ ] Verified access_30DAY.log is present
- [ ] Verified linux_s_30DAY.log is present
- [ ] Verified db_audit_30DAY.csv is present
- [ ] Verified products.csv is present

Task 2: Ingest Web Application Data

Overview

You will upload web application access logs that contain HTTP requests, response codes, and user session information.

Steps:

1. **Access Splunk Web Interface:**
   - Open your browser
   - Navigate to your Splunk instance: `http://localhost:8000`
   - Log in with username: `admin` and password: `password`
2. **Navigate to Add Data:**
   - Click **Settings** in the top menu
   - Select **Add Data**

   > **Note:** If you don't see the Add Data option, ensure you're logged in as admin
3. **Begin Upload Process:**
   - On the Add Data page, click **Upload**
   - Click **Select File**
   - Browse to your course materials `data` folder
   - Select `access_30DAY.log`
   - Click **Open**
4. **Review Source Type Detection:**
   - Splunk will analyze the file
   - Verify it automatically detected source type as `access_combined_wcookie`
   - Review the preview pane showing parsed events

- Click **Next**

5. **Configure Input Settings:**
   - **Host field value:** Enter `web_application`
   - Leave Index as `Default`
   - Click **Review**
6. **Review and Submit:**
   - Verify the following settings:
     - Input Type: Uploaded File
     - File Name: access_30DAY.log
     - Source Type: access_combined_wcookie
     - Host: web_application
     - Index: Default
   - Click **Submit**
7. **Verify Upload:**
   - Wait for the progress bar to complete
   - You should see "File has been uploaded successfully"
   - Click **Start Searching** to verify data
8. **Quick Validation Search:**

```
index=main host=web_application earliest=-30d
| stats count
```

   - Verify you see events from the uploaded file

Checkpoint ✓

- [ ] Successfully logged into Splunk
- [ ] File uploaded without errors
- [ ] Source type correctly identified
- [ ] Host field set to web_application
- [ ] Events visible in search

Task 3: Ingest Linux Secure Logs

Overview

Upload Linux secure logs containing authentication and security events from web servers.

Steps:

1. **Return to Add Data:**
   - Click **Add More Data** from the success screen
   - Or navigate: **Settings** → **Add Data** → **Upload**
2. **Select Linux Log File:**
   - Click **Select File**
   - Choose `linux_s_30DAY.log`
   - Click **Open**
3. **Configure Source Type:**
   - Notice Splunk shows source type as `default`
   - Click the **Source type** dropdown button
   - In the search box, type "linux_secure" or navigate to:
     - Expand **Operating System** category
     - Select **linux_secure**
   - Verify the preview updates with proper field extraction
   - Click **Next**
4. **Set Input Configuration:**
   - **Host field value:** Enter `web_server`
   - Keep Index as `Default`
   - Click **Review**
5. **Submit Configuration:**
   - Verify settings:
     - File Name: linux_s_30DAY.log
     - Source Type: linux_secure
     - Host: web_server
     - Index: Default
   - Click **Submit**

6. **Validation Search:**

```
index=main host=web_server sourcetype=linux_secure earliest=-30d
| head 10
```

Checkpoint ✓

- [ ] Source type manually configured
- [ ] Host field set to web_server
- [ ] Events properly parsed
- [ ] Authentication events visible

Task 4: Ingest Database Audit Logs

Overview

Upload CSV-formatted database audit logs and create a custom source type.

Steps:

1. **Start Upload Process:**
   - Click **Add More Data** from the success screen
   - Or navigate: **Settings** → **Add Data** → **Upload**
   - Click **Select File**
   - Choose `db_audit_30DAY.csv`
   - Click **Open**
2. **Create Custom Source Type:**
   - Splunk detects source type as `csv`
   - Click **Save As** button (next to source type)
   - In the modal window, enter:
     - **Name:** `db_audit`
     - **Description:** `Postgres Database Audit Logs`
     - **Category:** Select `Database`
     - **App:** Keep as `system`
   - Click **Save**
3. **Review Field Extraction:**
   - Verify the preview shows proper column headers

- Confirm fields are properly extracted
- Click **Next**

4. **Configure Input Settings:**
   - **Host field value:** Enter `database`
   - Index: `Default`
   - Click **Review**
5. **Final Submission:**
   - Verify all settings:
     - File Name: db_audit_30DAY.csv
     - Source Type: db_audit
     - Host: database
     - Index: Default
   - Click **Submit**
6. **Verify with Search:**

```
index=main sourcetype=db_audit host=database earliest=-30d
| table _time, Time, Type, Command, Duration
| head 10
```

Checkpoint ✓

- [ ] Custom source type created
- [ ] CSV fields properly extracted
- [ ] Host field set to database
- [ ] Tabular data displays correctly

Task 5: Verify All Data Sources

Overview

Confirm all three data sources are properly ingested and searchable.

Steps:

1. **Run Summary Search:**

```
index=main earliest=-30d latest=now
| stats count by sourcetype, host
| sort -count
```

2. **Expected Results:** You should see three rows:
   - sourcetype=access_combined_wcookie, host=web_application
   - sourcetype=linux_secure, host=web_server
   - sourcetype=db_audit, host=database
3. **Check Time Range Coverage:**

```
index=main earliest=-30d latest=now
| stats earliest(_time) as earliest, latest(_time) as latest by sourcetype
| eval earliest=strftime(earliest, "%Y-%m-%d %H:%M:%S")
| eval latest=strftime(latest, "%Y-%m-%d %H:%M:%S")
```

4. **Field Discovery Check:**

```
index=main sourcetype=access_combined_wcookie | head 1
```

   - Click on the event
   - Verify you can see extracted fields like status, clientip, method

Checkpoint ✓

- [ ] All three source types visible
- [ ] Event counts reasonable
- [ ] 30-day time range confirmed
- [ ] Fields properly extracted

Challenge Exercises

Challenge 1: Reconcile the Ingest

Confirm that what Splunk indexed matches what you uploaded. Count the events per source type and compare them against the line counts of the source files.

```
index=main
| stats count by sourcetype
```

Compare with `wc -l` on the three files in `labs/data/`.

Challenge 2: Confirm the Data Window

Determine how many days the dataset actually spans, rather than assuming.

```
index=main
| stats min(_time) as first, max(_time) as last
| eval span_days=round((last-first)/86400,1)
| eval first=strftime(first,"%Y-%m-%d"), last=strftime(last,"%Y-%m-%d")
| table first, last, span_days
```

Why does the course use **Last 30 days** rather than a range matching the span?

Challenge 3: Verify Host Assignment

Each source type should carry exactly one host value, set during upload. Prove it:

```
index=main
| stats dc(host) as distinct_hosts, values(host) as host_value by sourcetype
```

If any source type shows more than one host, the upload was misconfigured.

Challenge 4: Check Ingestion Health

Splunk logs its own activity. Look for problems recorded while your data loaded:

```
index=_internal sourcetype=splunkd log_level=ERROR
| stats count by component
| sort -count
```

Troubleshooting Guide

Common Issues and Solutions

Issue	Solution
"Add Data" option not visible	Log out and log back in with admin credentials
Source type not auto-detected	Manually select from the source type menu
No events in search	Check time range picker is set to "Last 30 days"
Fields not extracted in CSV	Ensure first row contains headers, re-upload if needed
Upload fails with size error	Check available disk space on Splunk server
Browser timeout during upload	Use Chrome/Firefox, disable aggressive timeout settings

Getting Help

1. Check the Splunk UI for error messages
2. Review `index=_internal` for ingestion errors
3. Verify file permissions and formats
4. Ask instructor for assistance

Lab Summary

What You Accomplished

✅ Located and verified sample data files
✅ Successfully uploaded three different data types
✅ Configured appropriate source types
✅ Set identifying host values
✅ Verified data ingestion through searches
✅ Created a custom source type for database audit logs

Key Concepts Covered

- **Data Ingestion Methods:** File upload through Splunk Web
- **Source Types:** Automatic detection and manual configuration
- **Host Field:** Identifying data sources
- **Index Selection:** Using default index for training
- **Data Verification:** Using SPL to confirm ingestion

Skills Developed

1. Navigating Splunk Web interface
2. Uploading data files
3. Configuring source types
4. Creating custom source types
5. Basic SPL search commands
6. Data validation techniques

Next Steps

In the next lab, you will:

- Learn basic search commands
- Work with time ranges
- Filter and transform the data you just loaded

- Create your first saved searches

End of Lab 1 | **Continue to Lab 2 →**

Clean-up Instructions

Note: Do NOT clean up data for this course as it will be used in subsequent labs.

If you need to reset your environment later:

1. Navigate to Settings → Indexes
2. Clean the main index events (instructor will demonstrate if needed)
3. Re-run this lab to reload data

Additional Notes

- The data loaded in this lab represents roughly 26 days of historical data
- These datasets will be used throughout the course
- Real-time data ingestion will be covered in advanced courses
- Keep note of the host values you assigned - you'll need them in future labs

Congratulations on completing Lab 1! 🎉